



B

esp32s3-sd

ESP32-S3, microSD, status LED

File: esp32s3-sd.kicad_sch

usb-uart

USB-C, CP2102N bridge

File: usb-uart.kicad_sch

power

Power: 5V OR, 3.3V buck, RP LDO

File: power.kicad_sch

C

D

E

F

12345678

FujiNet for the Atari 7800 – Rev0 (RP2354B). SCHEMATIC ONLY: no board laid out or built.

RP2354B (@200 MHz) sits on the console’s 5V cart bus through its 5V-tolerant pads, no buffers: D0–D7 (GPIO0–7) through 100R, A0–A15 (GPIO8–23), R/W, PHI2, /HALT (observed only) in; /IRQ through a 2N7002 (fitted, never asserted); PWM audio (GPIO29) through an RC into edge pin 18. It serves the boot block, the mailbox arena and POKEY itself and drives SRAM A13–A18, ROM_EN, RAM_EN and A8MASK (GPIO32–40) from a PIO slot table indexed by A13–A15. Pin map: fujinet-firmware pico/atari-7800 a78_cart.h. AS6C4008 (512K) on +5V: console A0–A7, A9–A12 and D0–D7 direct; SRAM A8 = A8 & !A8MASK. Four 74HCT packages make CARTSEL, SRAM /OE and /WE (a78_cartsel, a78_glue_oe, a78_glue_we, a78_glue_a8), checked gate by gate by tools/check_glue.py. PWR_OK (console +5V sense, 74HCT14) gates both. ESP32-S3-WROOM-1-N16R8 runs FujiNet (fujiversal-atari7800), USB host of the RP; S3 IO4/IO5 force RP RUN/QSPI_SS. USB-C: power + CP2102N UART bridge. microSD on S3 SPI. Power: console 5V via P-FET, USB via SS34. Edge: 32 pins, 2.54 mm, pins 1–16 on the console-rear face, pin 1 at the left (ASSUMED, verify on a real cart).

Generated by tools/gen_sch.py

FujiNet

Sheet: /

File: FujiNet-7800-Rev0.kicad_sch

Title: FujiNet 7800 Rev0

Size: A3

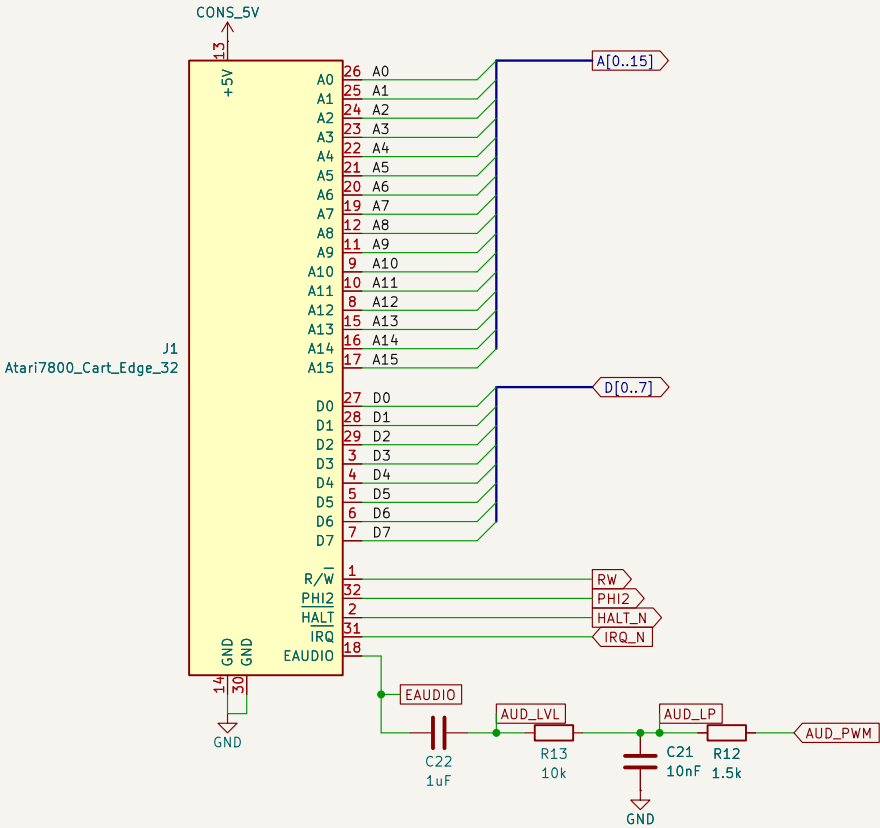
Date: 2026-10-06

Rev: 0

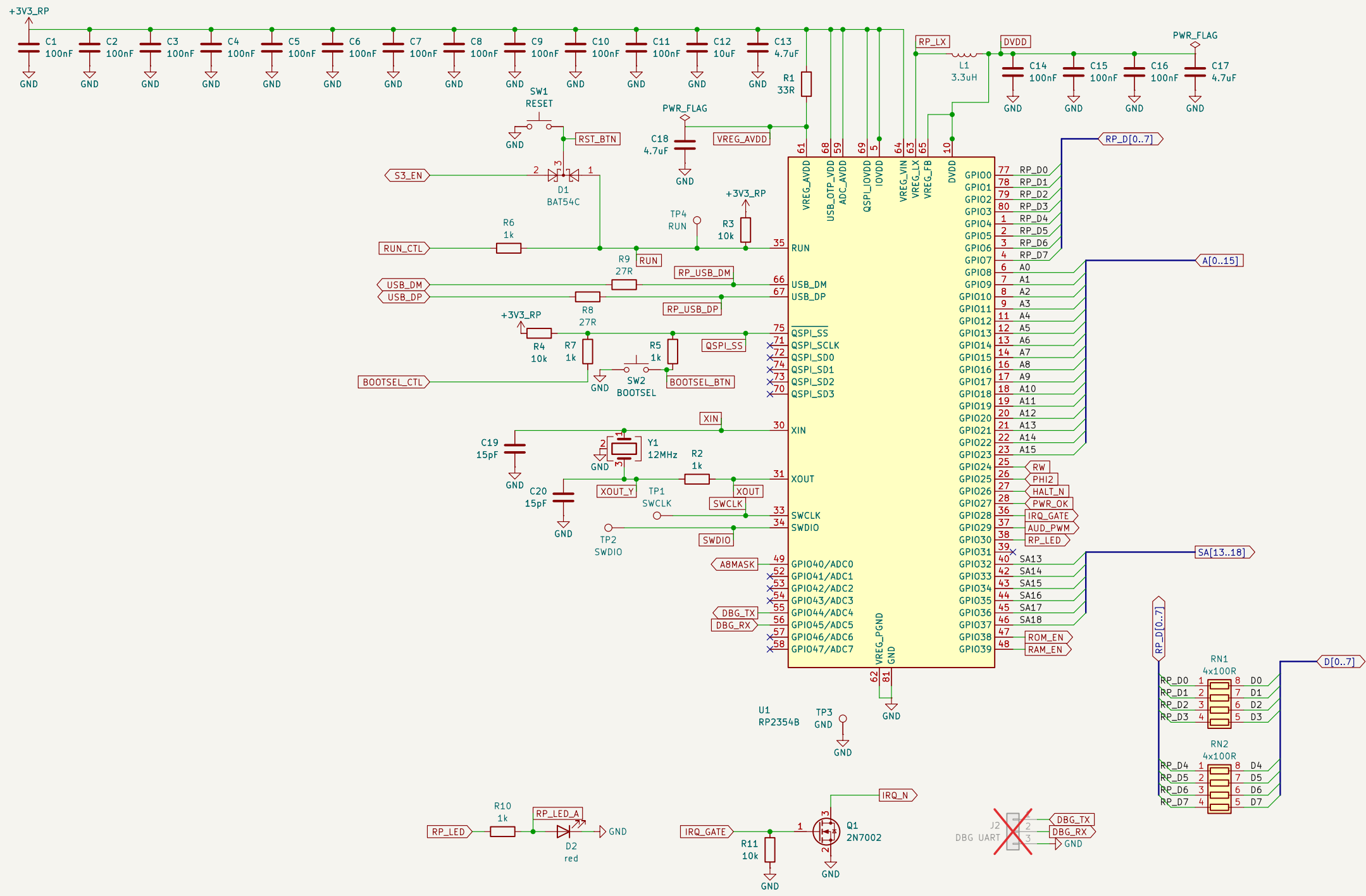
KiCad E.D.A. 10.0.6

Id: 1/8

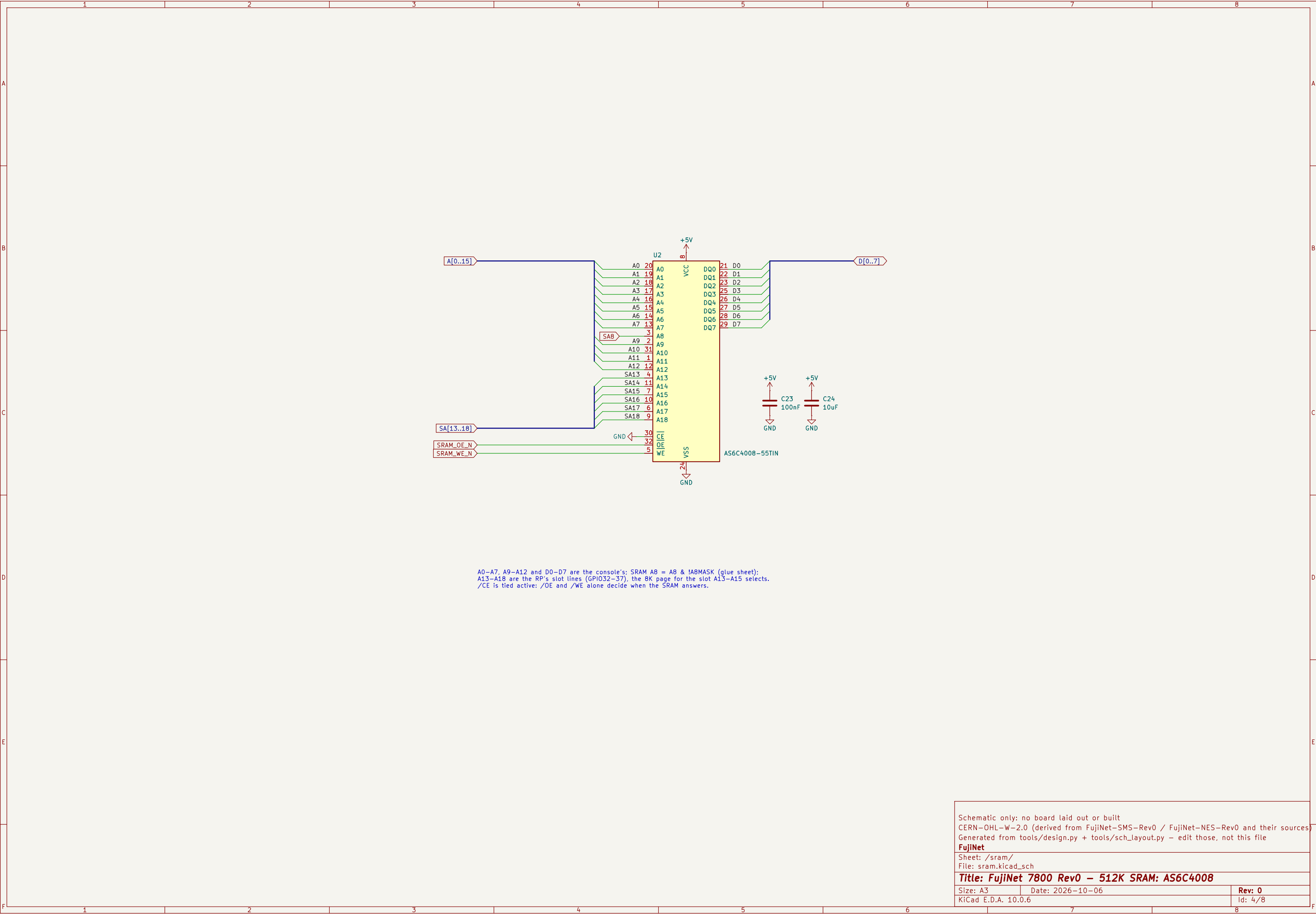
Atari 7800 32-pin cartridge edge, drawn by function (finger numbers on the pins).
Pins 1-16 face pins 32-17 (k over 33-k); 3-14 and 19-30 are the 2600 edge. ASSUMED, verify on a real cart PCB: pins 1-16 on the console-rear face, pin 1 at the LEFT seen from that face with the fingers down. No chip select, no reset: the glue decodes A15-A11 itself.



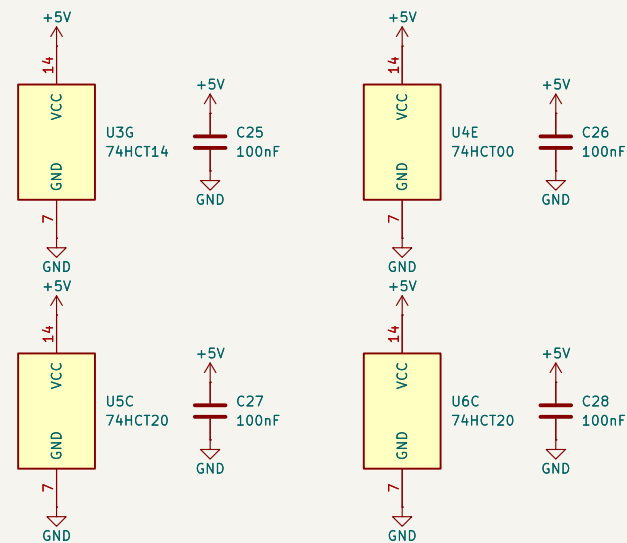
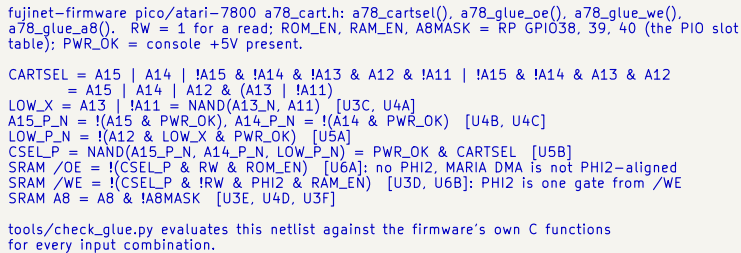
Cart audio: GPIO29 PWM (833 kHz carrier) -> 1.5k / 10n (fc 10.6 kHz) -> 10k -> 1u -> EAUDIO.
The 10k level is PROVISIONAL (match a POKEY cart on a real console); the 1u keeps the console's audio node at its no-cart DC level.



Activity LED (GPIO30). Console /IRQ: the 2N7002 pulls it low only while GPIO28 is high, which the firmware never does; the 10k holds the gate low from power-on.
RP debug UART J2 (DNP, 3.3 V only): 1 TX GPIO44, 2 RX GPIO45, 3 GND.



A0–A7, A9–A12 and D0–D7 are the console's; SRAM A8 = A8 & !A8MASK (glue sheet);
A13–A18 are the RP's slot lines (GPIO32–37), the 8K page for the slot A13–A15 selects.
/CE is tied active: /OE and /WE alone decide when the SRAM answers.



Schematic only: no board laid out or built		
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources)		
Generated from tools/design.py + tools/sch_layout.py – edit those, not this file		
FujiNet		
Sheet: /glue/		
File: glue.kicad_sch		
Title: FujiNet 7800 Rev0 – 5V glue: CARTSEL, SRAM /OE, /WE, A8, PWR_OK		
Size: A3	Date: 2026-10-06	Rev: 0
KiCad E.D.A. 10.0.6		Id: 5/8

